Lecture 7

Semiconductors
Diodes and its Applications

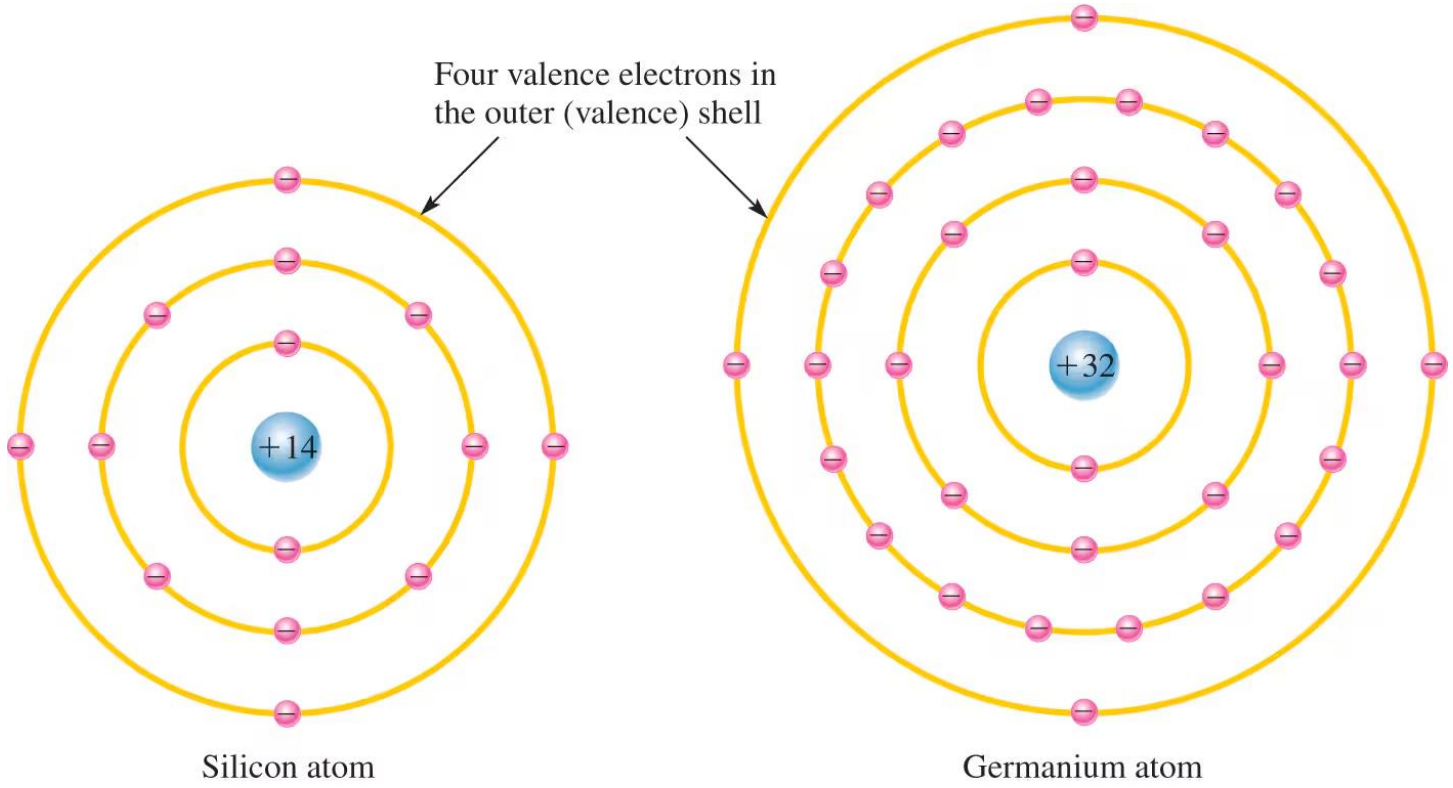
Slide Courtesy:
1.Electronics Technology Fundamentals, 3rd ed. Paynter and Boydell
2. Electronics Fundamentals 9th edition, Floyd/Buchla/Snyder

Outline

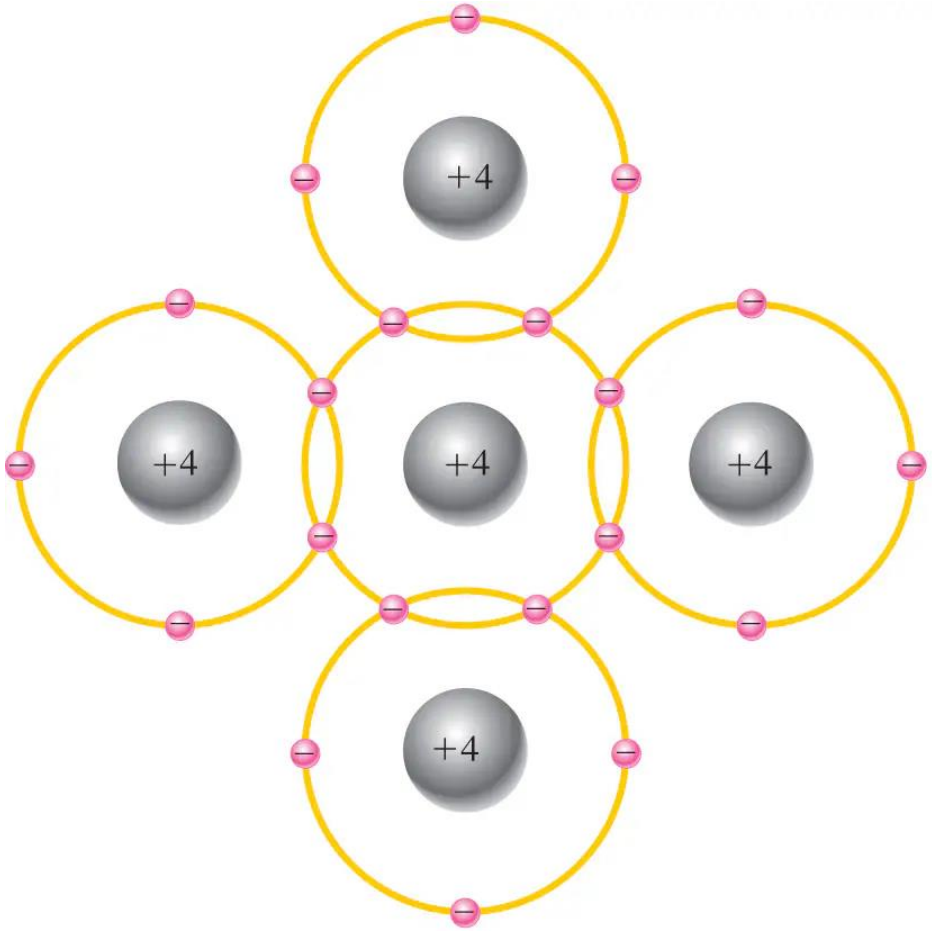
1. Semiconductors
   - Atomic Bonding in Semiconductors
   - Intrinsic Semiconductors
   - Doping and Extrinsic Semiconductors
   - PN Junction Theory
   - Forward and Reverse Bias of PN Junction
2. Diode
3. LED – Light Emitting Diode
4. Zener Diode
5. Diode Application
   - Rectifiers

Atomic Bonding in Semi Conductors

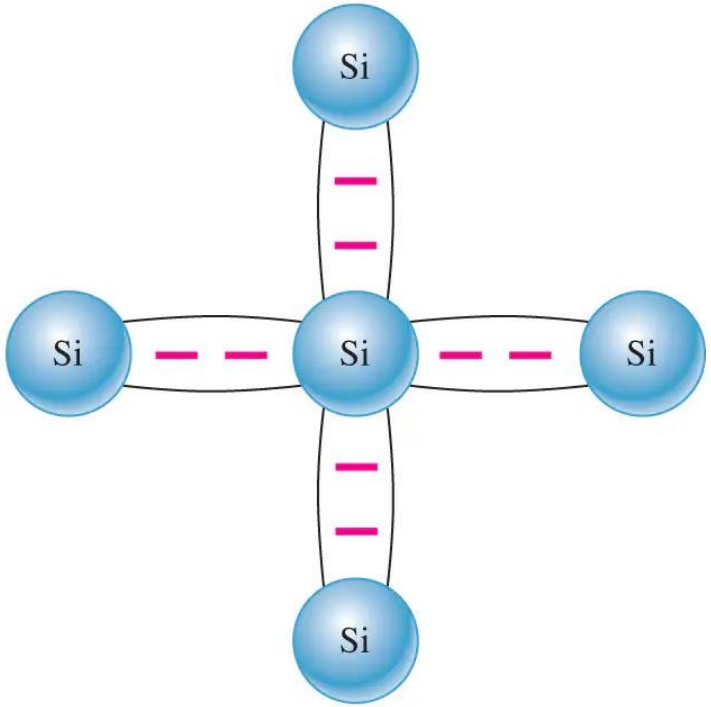
- Two types of semiconductor materials are **Silicon (Si)** and **Germanium (Ge)**.
- Silicon (Si) and germanium (Ge), have **four valence electrons** in their outer shell. These valence electrons are shared between neighboring atoms, creating a stable structure.
- The outermost electrons, or valence electrons, play a critical role in the behavior of materials, especially in conductivity.



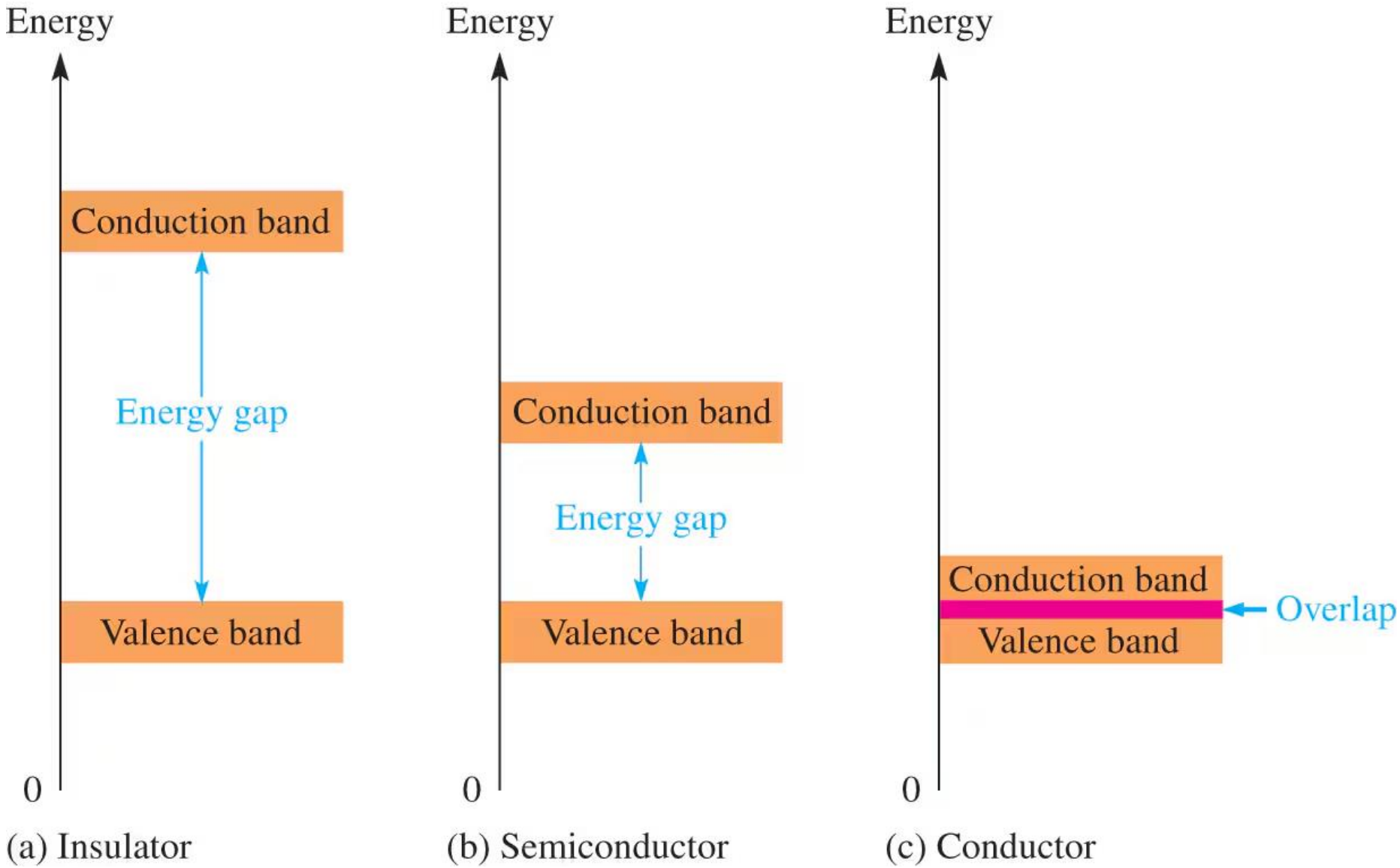
Covalent Bonding in Semiconductors

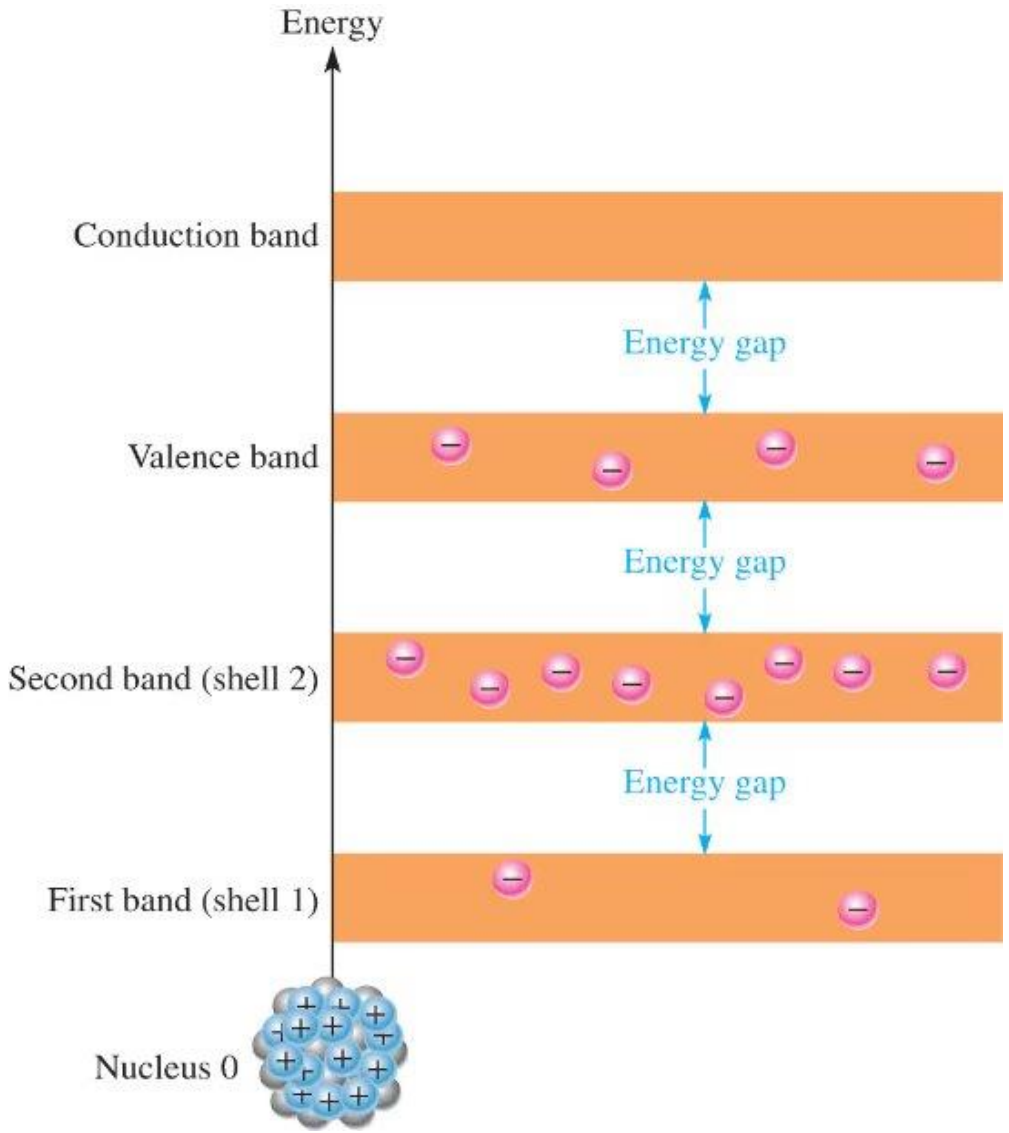


(a) The center silicon atom shares an electron with each of the four surrounding silicon atoms, creating a covalent bond with each. The surrounding atoms are in turn bonded to other atoms, and so on.



(b) Bonding diagram. The red negative signs represent the shared valence electrons.


Energy
Conduction band
Energy gap
Valence band
0
(a) Insulator
Energy
Conduction band
Energy gap
Valence band
0
(b) Semiconductor
Energy
Conduction band
Overlap
Valence band
0
(c) Conductor

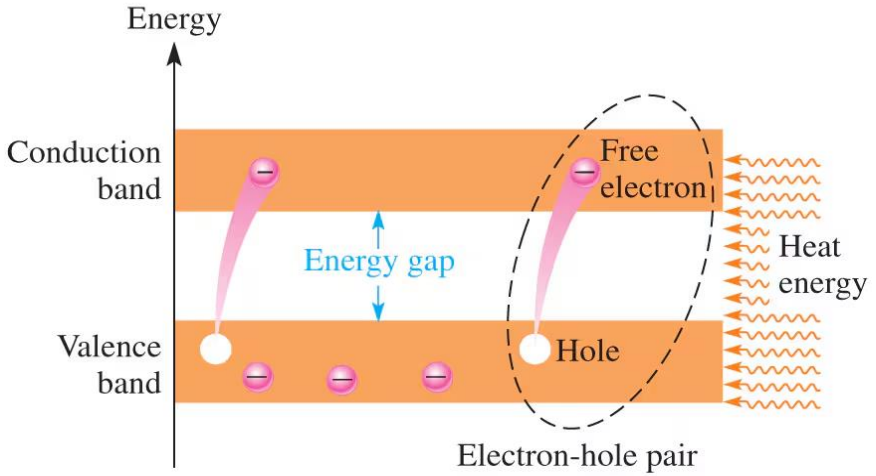


Energy band diagram for a pure silicon crystal with unexcited atoms. There are no electrons in the conduction band.

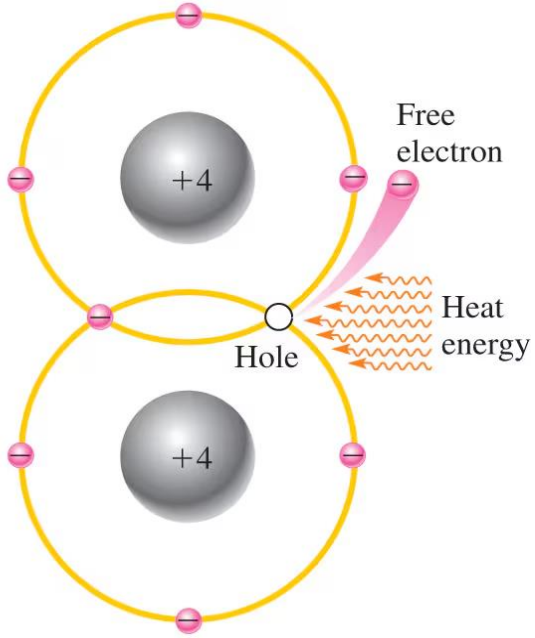
- **Why Covalent Bonding is Important in Semiconductors?**
- The covalent bonds in a semiconductor are what give it the ability to behave differently from **conductors** and **insulators**.
- At absolute **zero temperature**, semiconductors act like **insulators** because all electrons are bound in covalent bonds.
- As temperature increases, some electrons gain enough energy to break free from their covalent bonds, becoming **conduction electrons**, which can carry current.
- This **ability to control conductivity** is the foundation of how semiconductors work in devices like transistors and diodes.

Electron and Hole Current

- **Electron-Hole Pair Generation**: At room temperature, thermal energy generates free electrons and holes.
- **Free electron current** in intrinsic silicon is produced by the movement of thermally generated free electrons in the conduction band.
- **Hole Movement (Hole Current):** When an electron jumps to the conduction band, a vacancy is left in the valence band within the crystal. This vacancy is called a **hole.** As neighboring electrons "jump" to fill the hole left behind, this movement of electrons creates the **appearance** that the hole is moving in the opposite direction. This current is called ***hole current.***
- Carrier Concentration: Equal number of electrons and holes in intrinsic semiconductors.
  - Very few in number
  - Hard to control

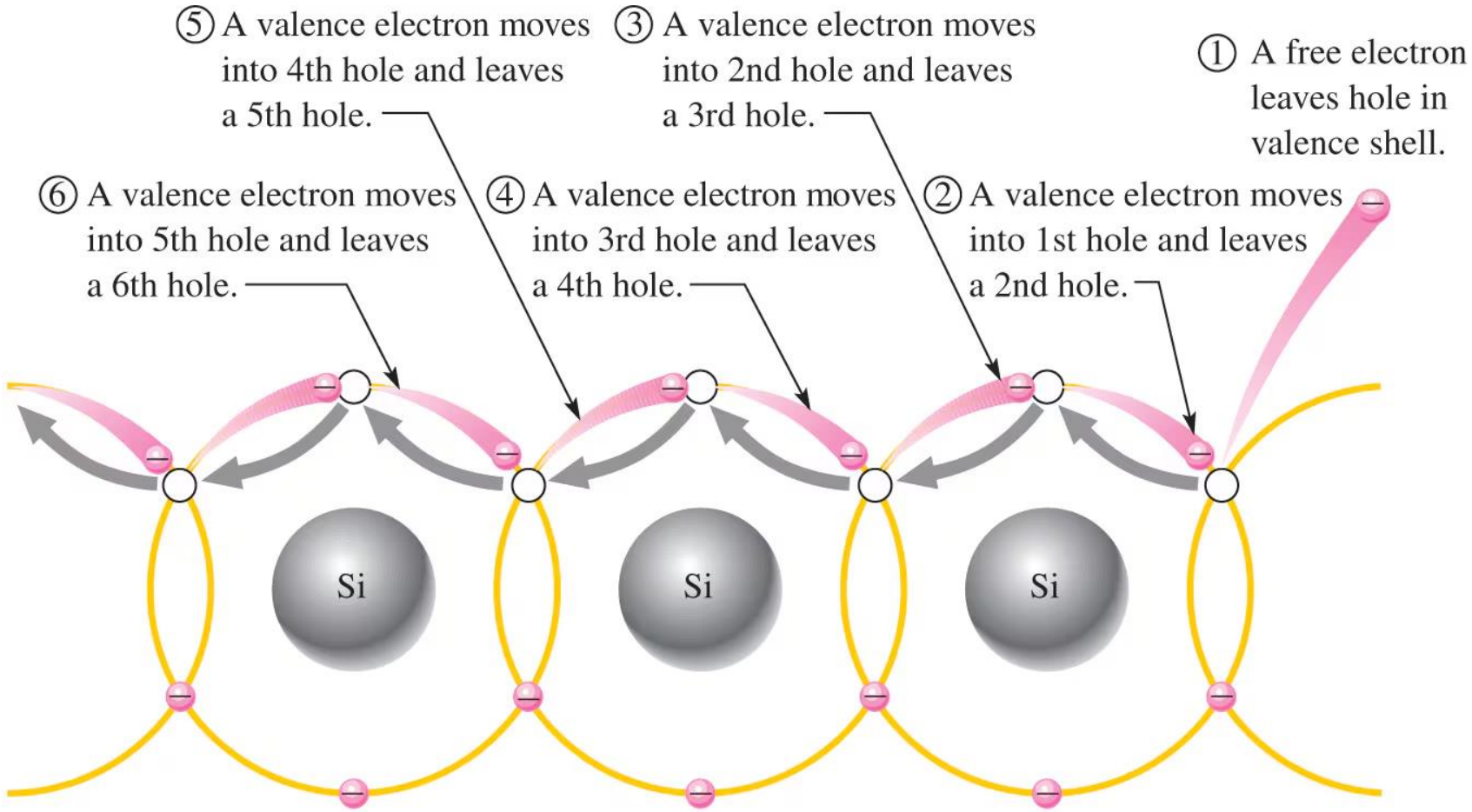


(a) Energy diagram



(b) Bonding diagram

Electron- Hole current



When a valence electron moves left to right to fill a hole while leaving another hole behind, the hole has effectively moved from right to left. Gray arrows indicate effective movement of a hole.

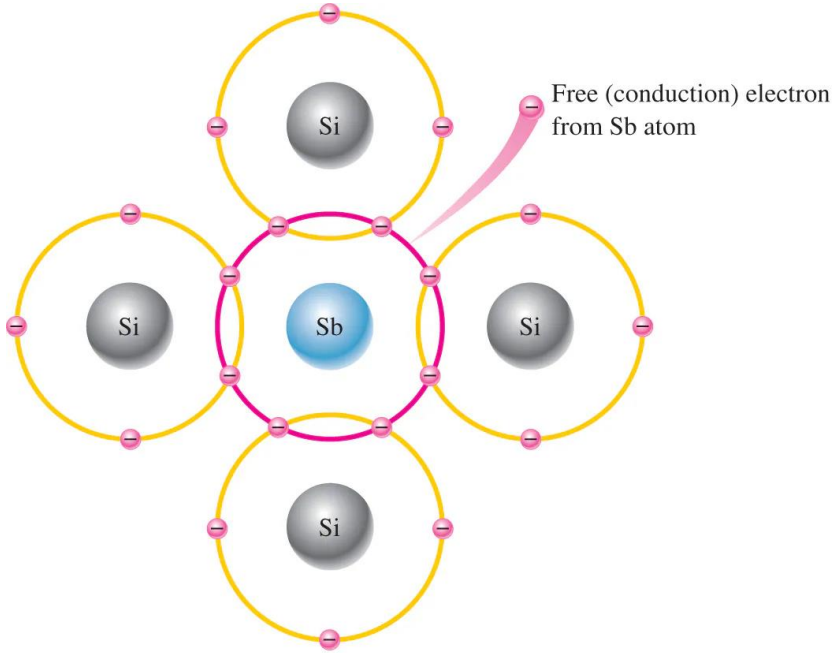
Doping and Extrinsic Semi Conductors

- Semiconductor in its **pure form – intrinsic semiconductor materia**l: No impurities, only silicon atoms.
- By adding certain impurities to pure (intrinsic) silicon, more holes or more electrons can be produced within the crystal.
- **Doping** – the process of adding impurity atoms to intrinsic silicon to increase its conductivity. The impure semiconductor is called **Extrinsic.**
- To increase the number of conduction band electrons, pentavalent impurities are added, forming an **n-type semiconductor**. These are elements to the right of Si on the Periodic Table.
- To increase the number of holes, trivalent impurities are added, forming a ***p*-type semiconductor**. These are elements to the left of Si on the Periodic Table.

III	IV	V
B	C	N
Al	Si	P
Ga	Ge	As
In	Sn	Sb

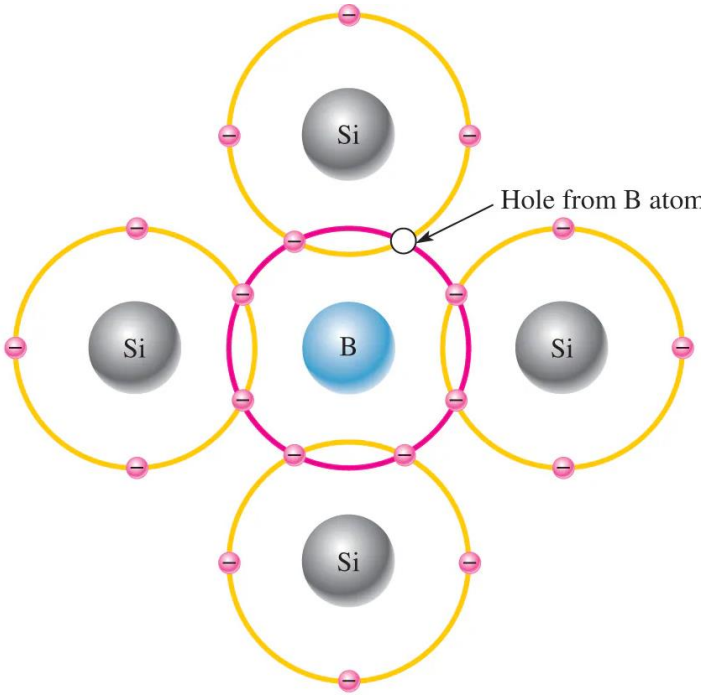
Extrinsic Semiconductors

- ***N*-Type Materials**
  - Add pentavalent impurities – resulting in excess electrons
  - Electrons are majority carriers (-ve charge)
  - Holes – minority carriers



(a) Pentavalent impurity atom in a silicon crystal. An antimony (Sb) impurity atom is shown in the center. The extra electron from the Sb atom becomes a free electron.

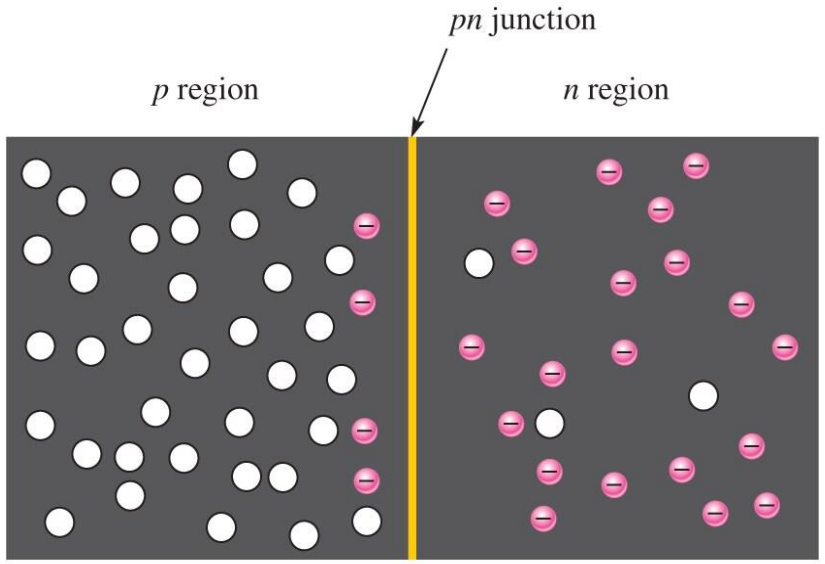
- ***P*-Type Materials**
  - Add trivalent impurities – resulting in absence of electrons
  - Holes are majority carriers (+ve charge)
  - Electrons – minority carriers



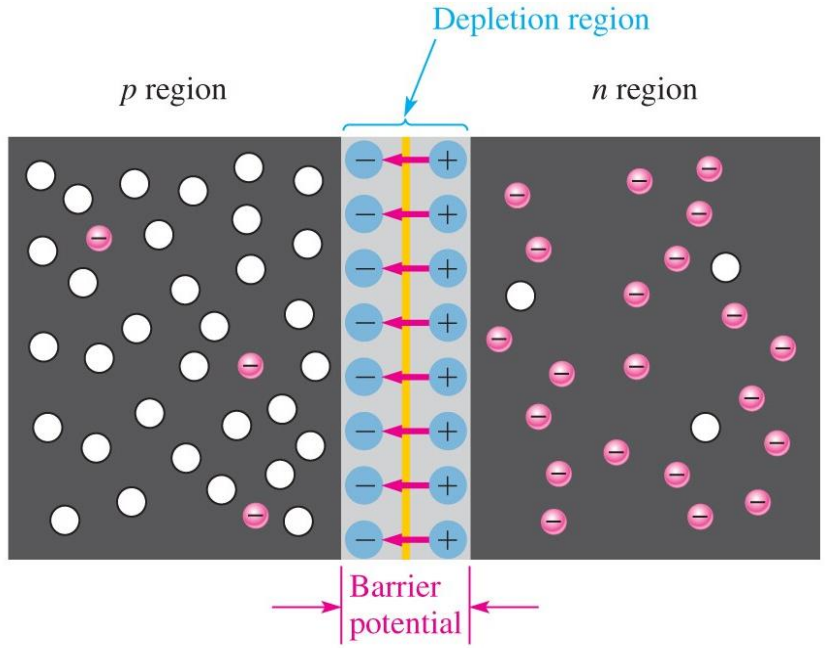
(b) Trivalent impurity atom in a silicon crystal. A boron (B) impurity atom is shown in the center.

The PN Junction Theory

- **P-N Junction:** Formed by joining P-type and N-type materials.
- **Depletion Region:** Area around the junction where free carriers (electrons and holes) recombine, creating an electric field.
- **Built-in Potential:** Voltage that develops across the junction due to the separation of charges.



(a) At the instant of junction formation, free electrons in the *n* region near the *pn* junction begin to diffuse across the junction and fall into holes near the junction in the *p* region.



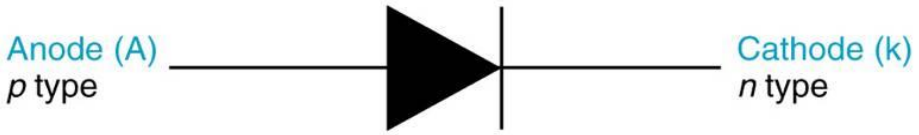
(b) For every electron that diffuses across the junction and combines with a hole, a positive charge is left in the *n* region and a negative charge is created in the *p* region, forming a barrier potential. This action continues until the voltage of the barrier repels further diffusion.

- The **barrier potential** **(V_B)**,is the amount of voltage required to move electrons through the depletion region.
- At 25°C, it is approximately **0.7 V for silicon** and **0.3 V for germanium.**

PN Junction Diodes

- The PN junction is basically a diode.
- A diode is a fundamental semiconductor device commonly found in electronic circuits.
- Its primary function is to allow the flow of electric current in one direction while blocking it in the opposite direction.

Diode symbol



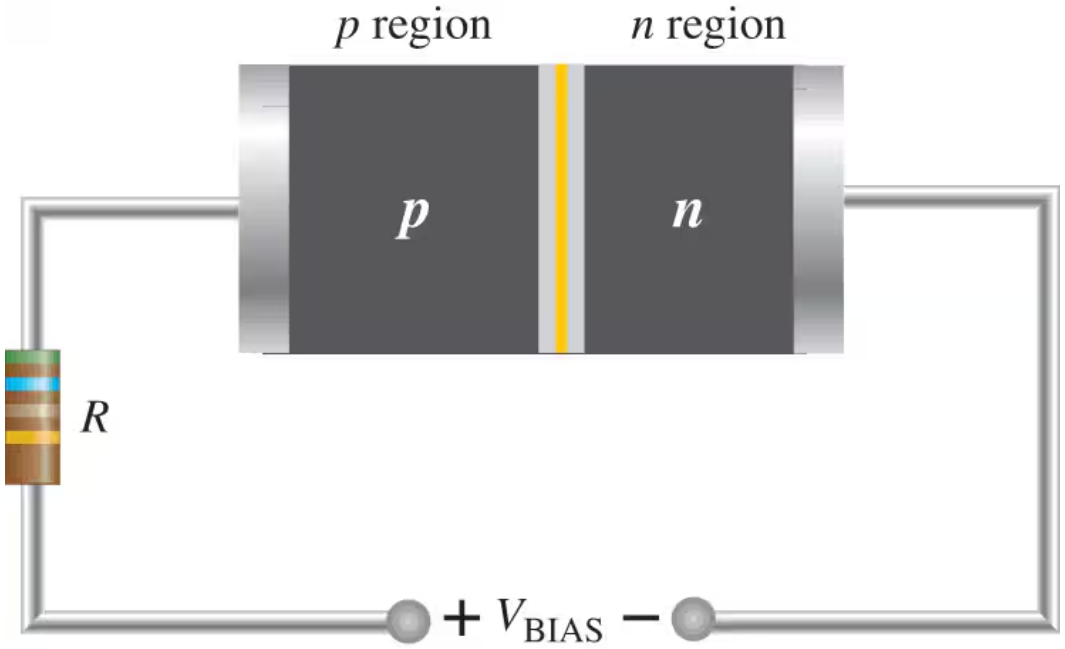
Forward and Reverse Bias of PN Junction

- Bias – a potential applied to a *pn* junction to control the width of its depletion layer

  Zero Bias:
  - No external voltage is applied to the P-N junction diode.

  Forward Bias:
  - External voltage reduces the depletion region.
  - Electrons and holes recombine, allowing current flow.

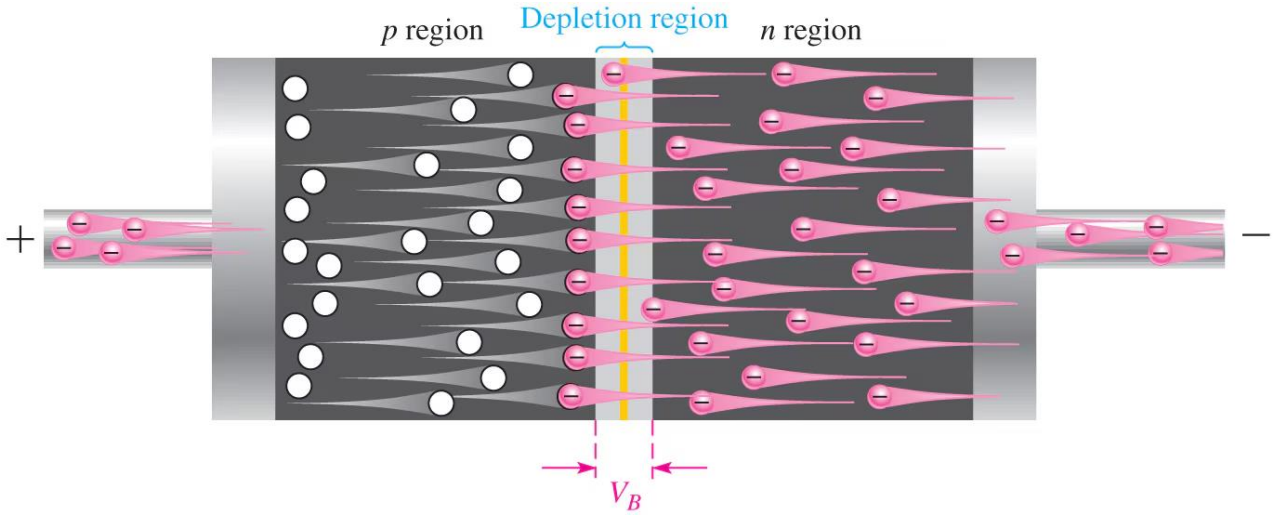
  Reverse Bias:
  - External voltage increases the depletion region.
  - No significant current flow (except leakage current).



Forward Bias

Forward Bias: When a *pn* junction is forward-biased, current is permitted. The bias voltage pushes conduction-band electrons in the *n*-region and holes in the *p*-region toward the junction where they combine.

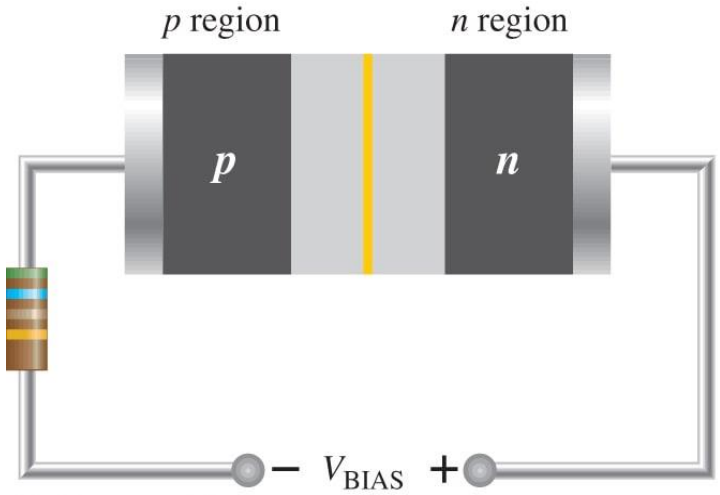
- Forward Bias – conduction happens
  - $V_B \cong 0.7$ V for silicon
  - $V_B \cong 0.3$ V for germanium
  - R_B is minimum



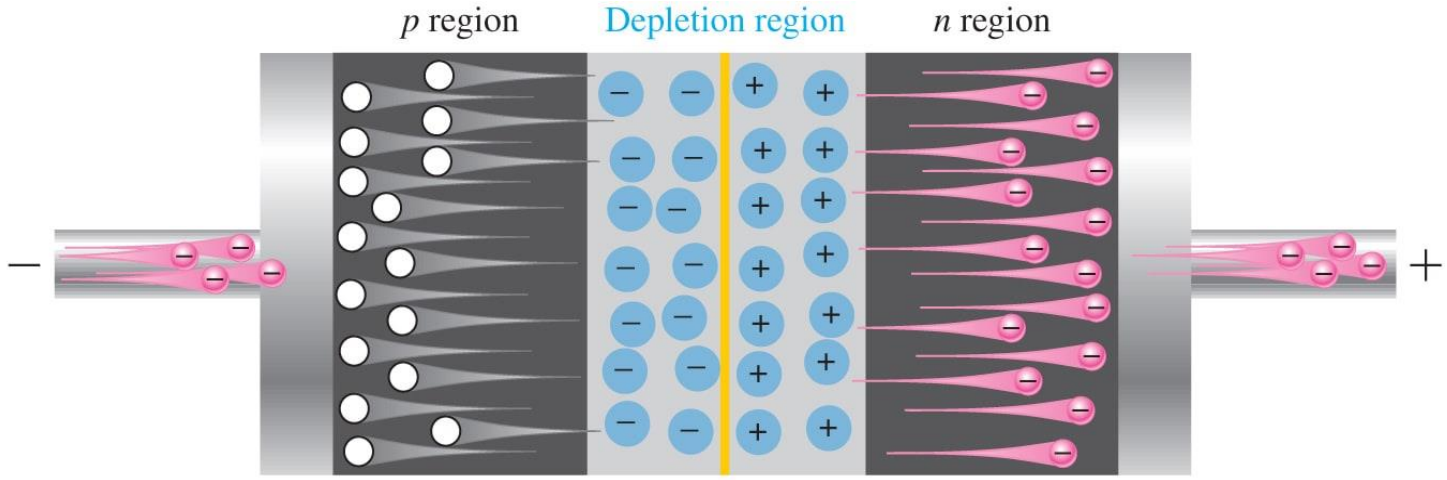
Forward bias → ON switch → conduction of current

Reverse Bias

- When a *pn* junction is reverse-biased, the bias voltage moves conduction-band electrons and holes away from the junction, so current is prevented.
- The diode effectively acts as an insulator. A relatively few electrons manage to diffuse across the junction, creating only a tiny reverse current.
- The reverse-bias causes the depletion region to widen.
- R_B is very high (infinite/OL) – no conduction



(a) Reverse-bias connection.

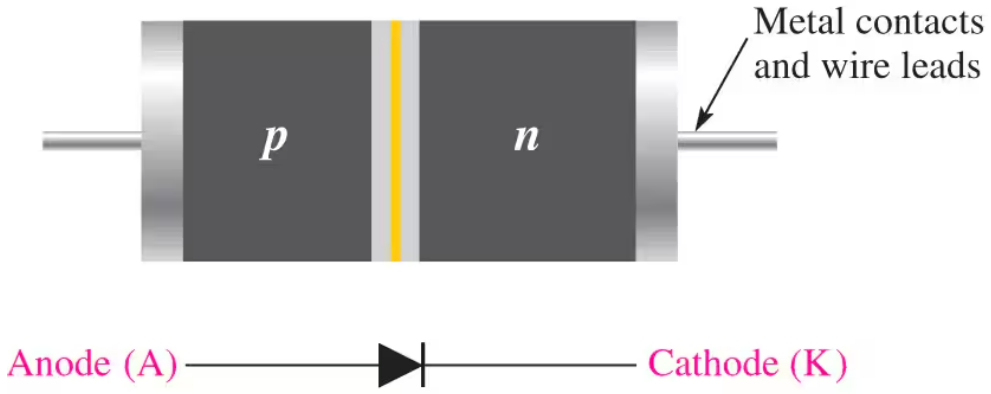


(b) There is transient current as depletion region widens.

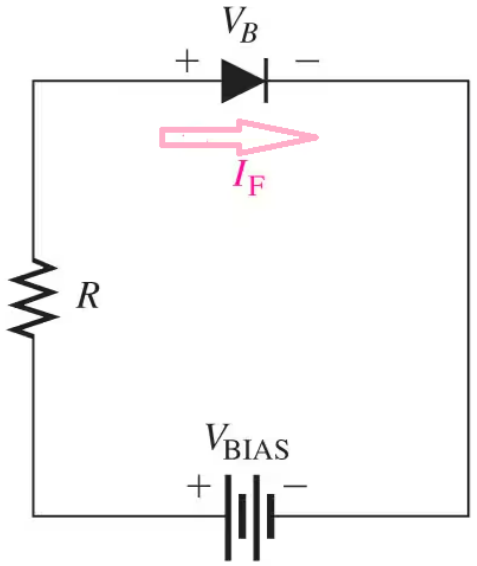
Reverse bias → OFF switch → No current flow.

PN Junction Diodes

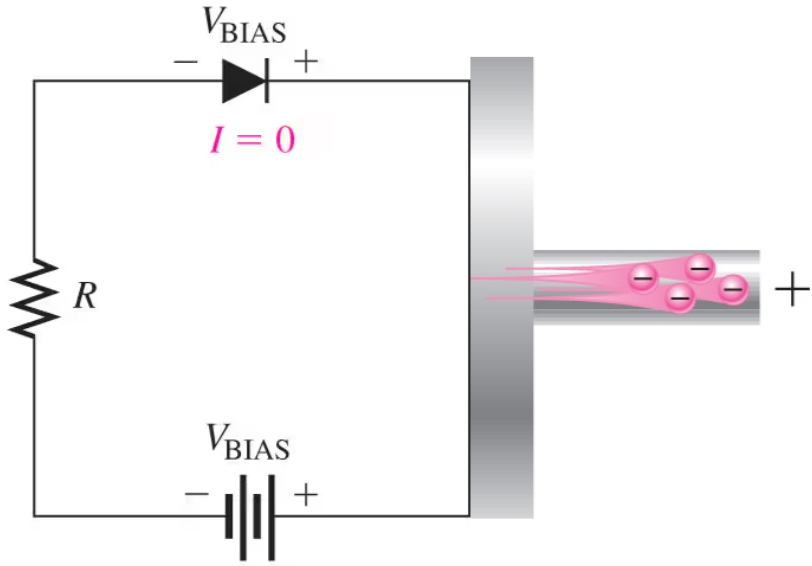
- Ideal Diode Characteristics – would act as a simple switch
  - Reverse Biased (Open Switch) – has infinite resistance, and zero reverse current.
  - Forward Biased (Closed Switch) – has no resistance, and allows current flow



(a) Basic diode structure and symbol



(b) Forward bias



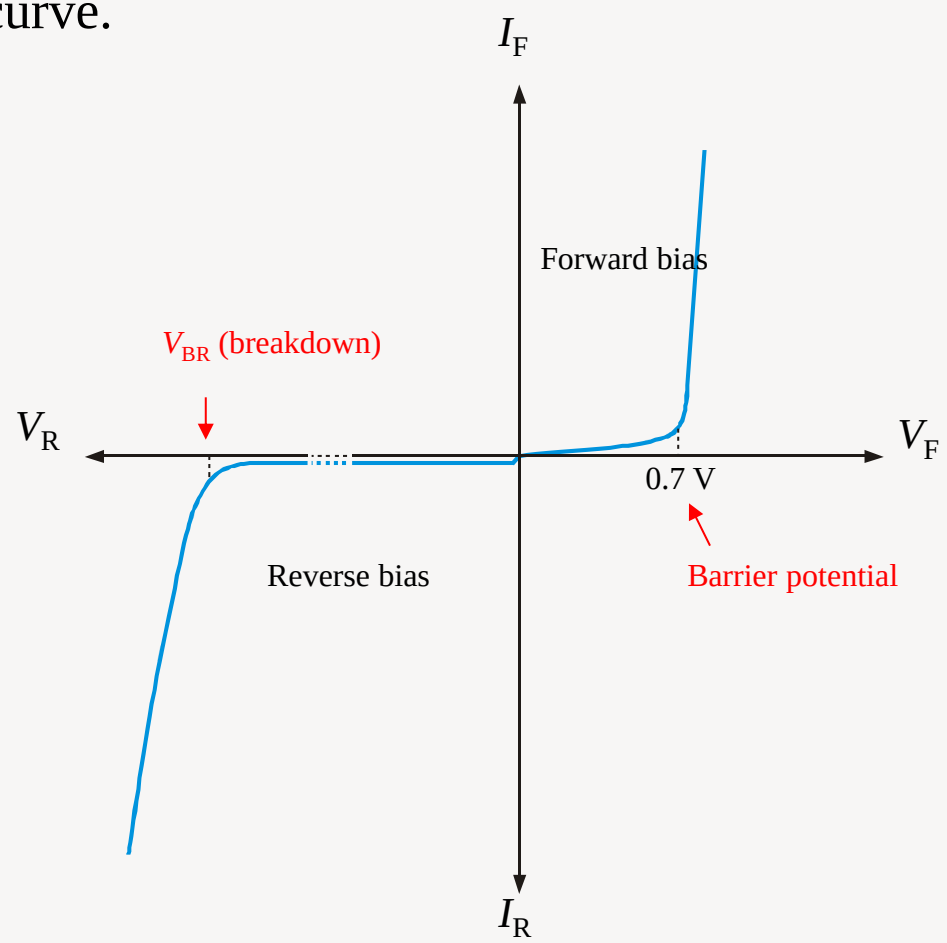
(c) Reverse bias

Diode Characteristics

- In the forward bias region, current increases dramatically after the barrier potential (0.7 V for Si) is reached. The voltage across the diode remains approximately equal to the barrier potential.

- The reverse-biased diode effectively acts as an insulator until breakdown is reached.

The forward and reverse characteristics are shown on a *V-I* characteristic curve.



Diode Types



Generic Diode



Zener Diode



Schottky Diode



LED



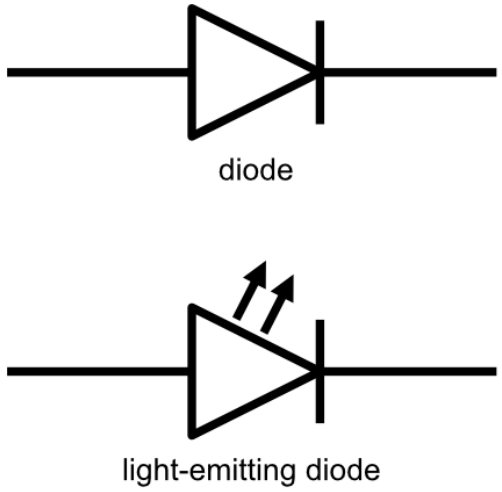
Tunnel Diode

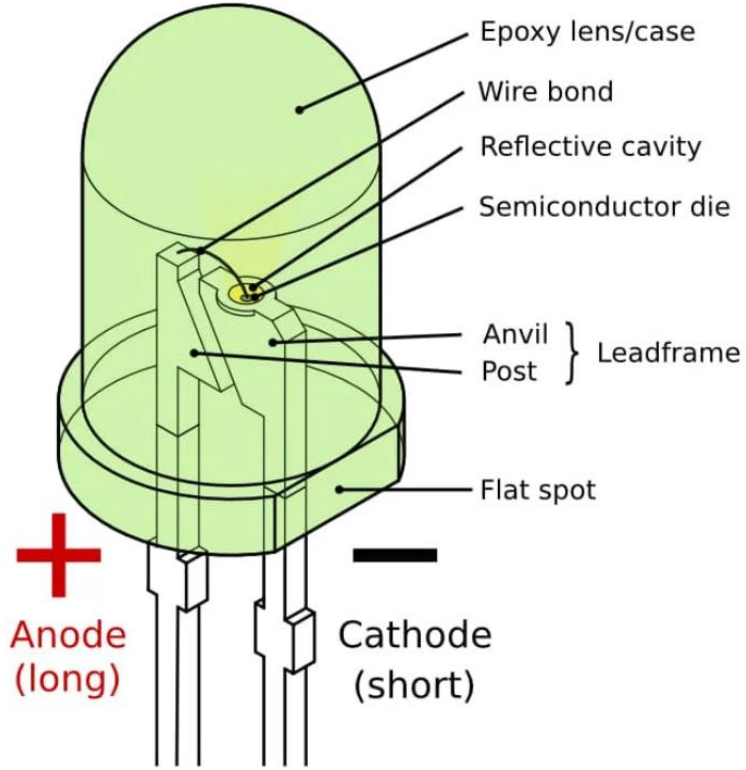


Photo Diode

LED – Light Emitting Diode

- An LED (Light Emitting Diode) is a special type of **diode** that lights up when current flows through it in the right direction.

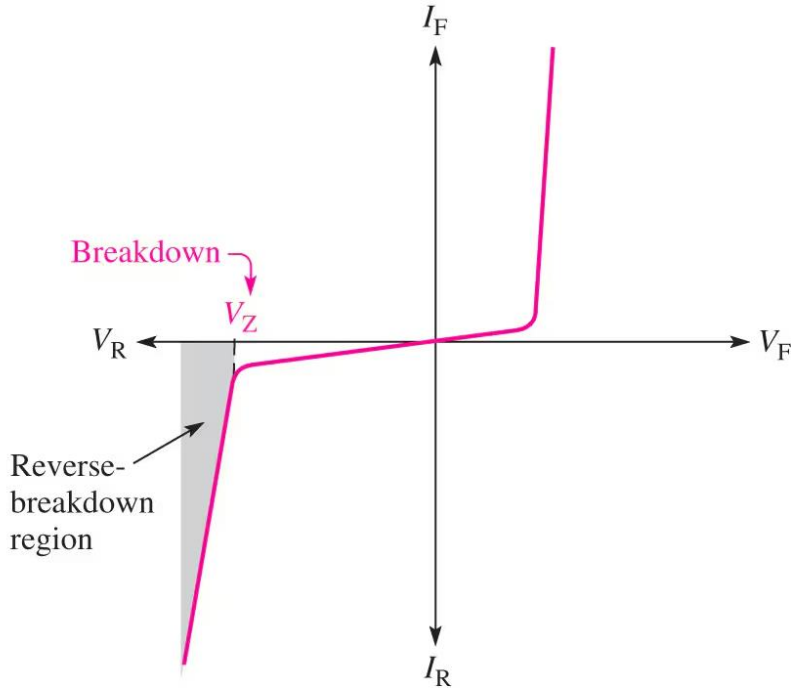




Zener Diode

- A Zener diode is a highly doped semiconductor device specifically designed to function in the **reverse direction.**
- The voltage at which a reverse biased Zener diode conducts current – **Zener Breakdown**
- If a zener diode is forward-biased, it operates the same as a rectifier diode..

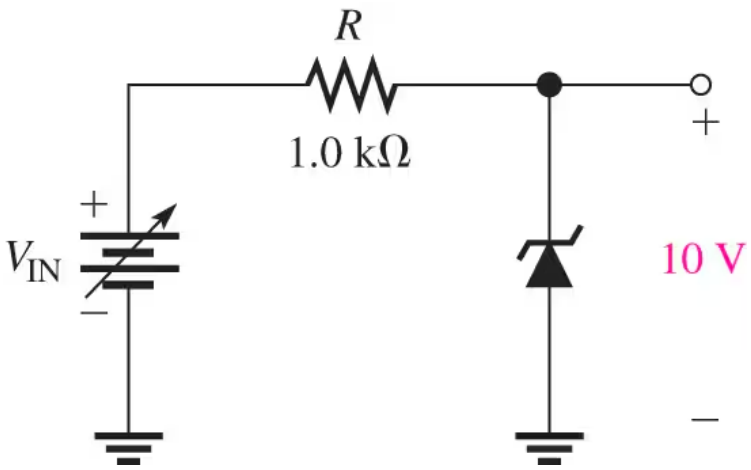




(b) The normal operating region for a zener diode is shaded.

Zener Diode as a Voltage Regulator

- Major application for the zener diode is to provide an output reference voltage that is stable despite changes in input voltage.
- As the input voltage varies (within limits), the zener diode maintains an essentially constant voltage across the output terminals.

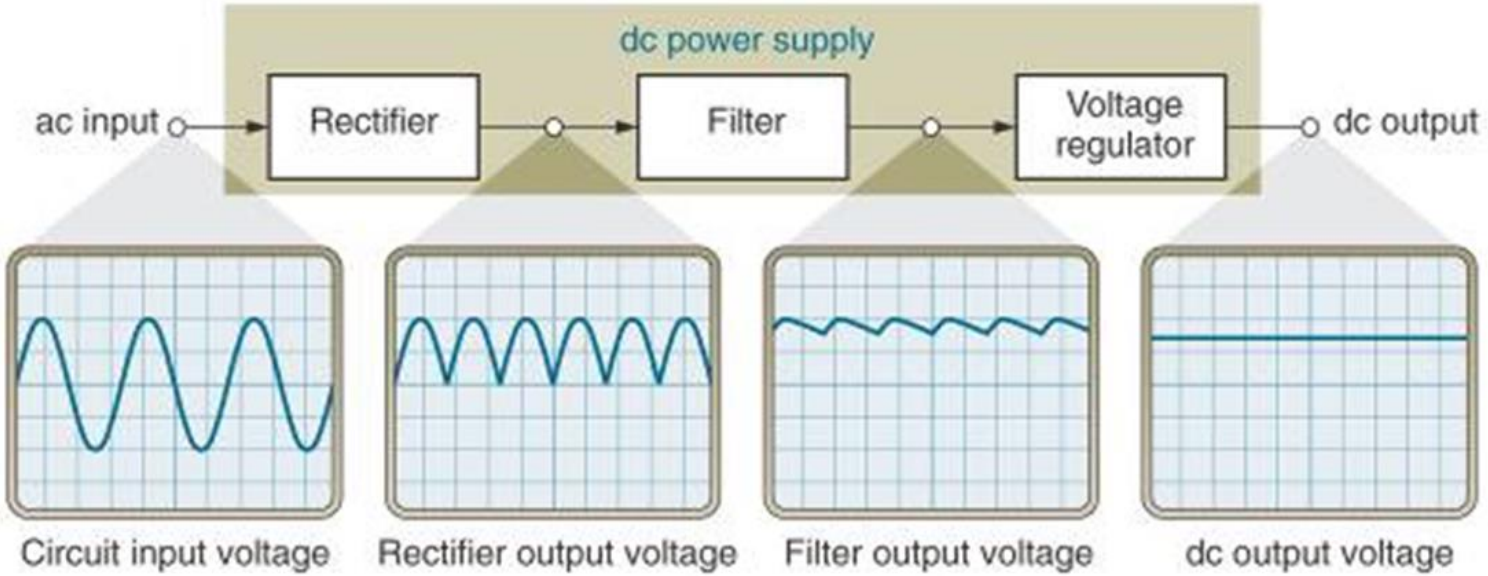


Basic Diode Circuits

Rectifier

Diodes are used in circuits called rectifiers that convert ac voltage into dc voltage.
Rectifiers are found in all dc power supplies that operate from an ac voltage source.

The Basic DC Power Supply



Basic power supply block diagram and waveforms

(a)



Basic power supply with a transformer input

(b)

Rectifier – a diode circuit that converts ac to pulsating dc

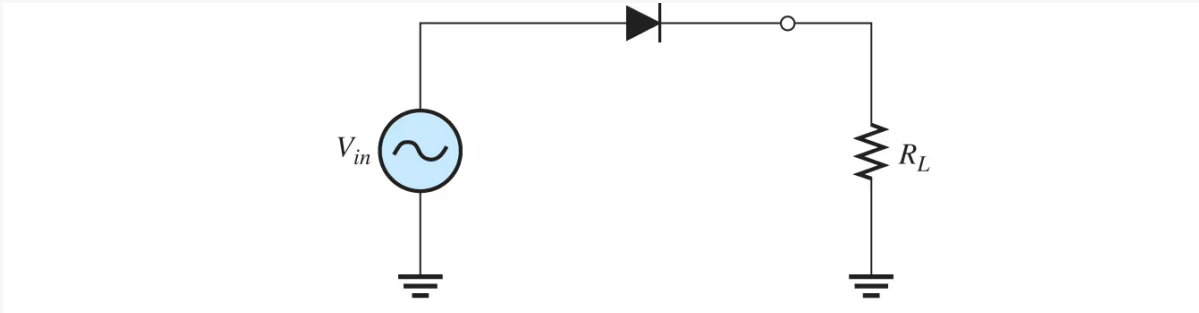
Filter – a circuit that reduces the variations in the output from the rectifier

Voltage Regulator – a circuit designed to maintain a constant power supply output voltage

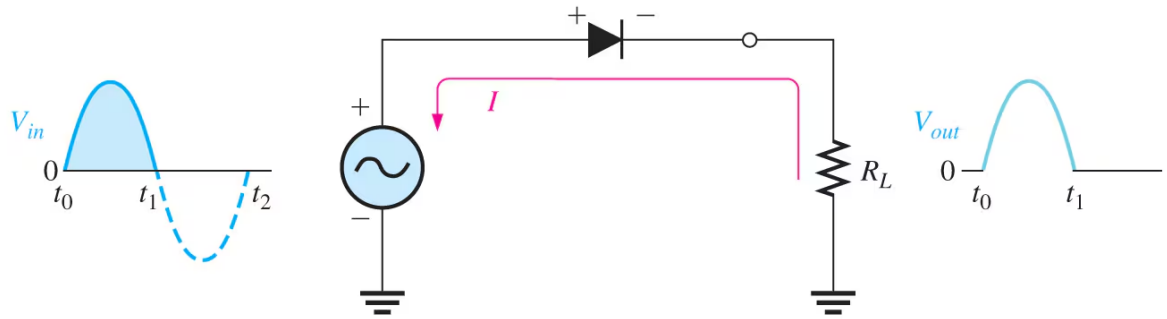
Half-wave Rectifier

The half-wave rectifier converts ac to pulsating dc by acting as a closed switch during the positive alteration.

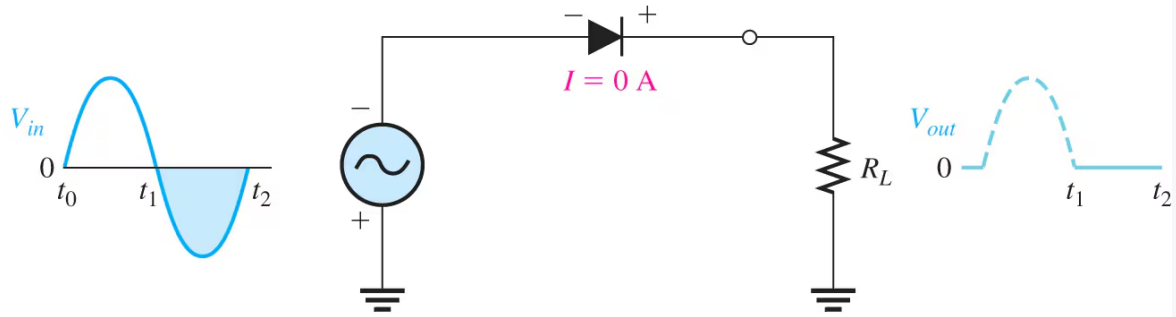
The diode acts as an open switch during the negative alteration.



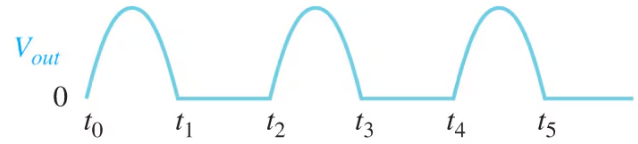
(a) Half-wave rectifier circuit



(b) Operation during positive alternation of the input voltage



(c) Operation during negative alternation of the input voltage

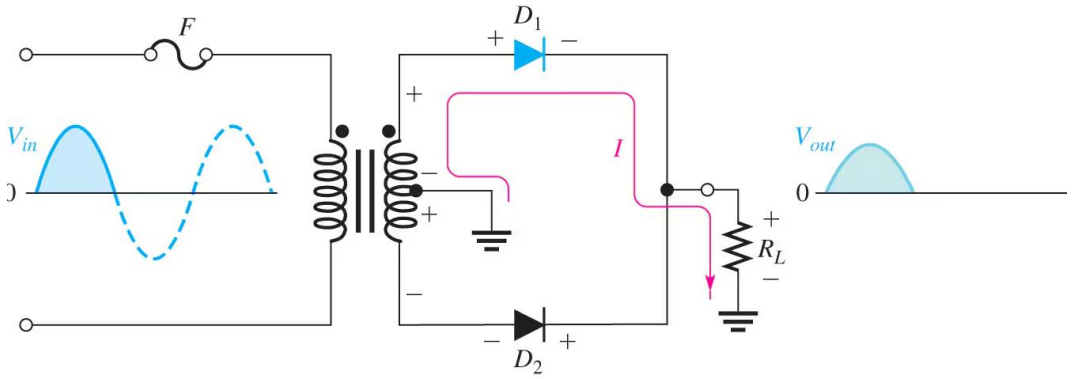


(d) Half-wave output voltage for three input cycles

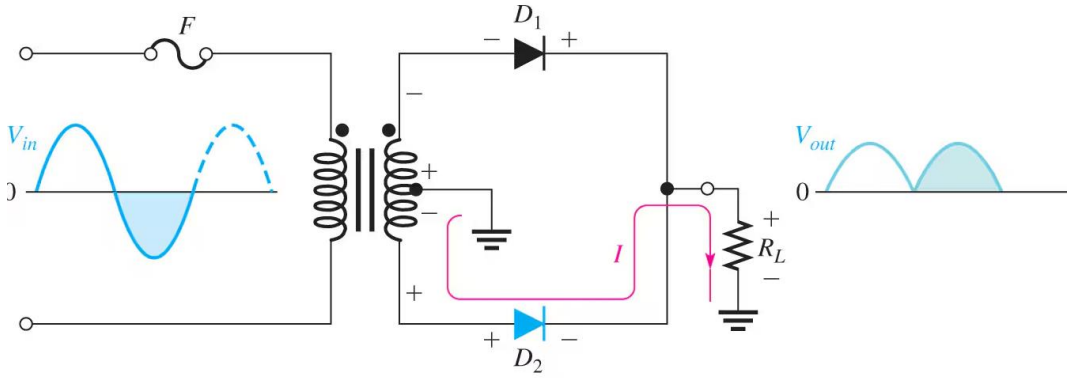
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Full-wave Rectifier

- The full-wave rectifier allows unidirectional current on both alterations of the input. The center-tapped full-wave rectifier uses two diodes and a center-tapped transformer.
- The ac on each side of the center-tap is ½ of the total secondary voltage. Only one diode will be biased on at a time.
- For a **positive half-cycle** of the input voltage, upper diode D1 is forward biased and lower diode D2 is reverse-biases the lower diode . The current path is through D1 and the load resistor RL
- For a **negative half-cycle** of the input voltage, D2 is forward biased and D1 is reverse biased. The current path is through D2 and RL, as indicated.
- Because the current during both the positive and the negative portions of the input cycle is in the same direction through the load, the output voltage developed across the load is a full-wave rectified dc voltage.



(a) During positive half-cycles, D_1 is forward-biased and D_2 is reverse-biased.



(b) During negative half-cycles, D_2 is forward-biased and D_1 is reverse-biased.

Bridge Rectifier

The bridge rectifier is a type of full-wave circuit that uses four diodes. The bridge rectifier does not require a center-tapped transformer.

At any instant, two of the diodes are conducting and two are off.

- When the input cycle is positive as in part (a), diodes D1 and D2 are forward-biased and conduct current in the direction shown. A voltage is developed across that looks like the positive half of the input cycle. During this time, diodes D3 and D4 are reverse-biased.
- When the input cycle is negative, diodes D3 and D4 are forward-biased and conduct current in the same direction through as during the positive half-cycle. During the negative half-cycle, D1 and D2 are reverse-biased. A full-wave rectified output voltage appears across as a result of this action.

